

Solving the Heat Equation by Separation of Variables

A conceptual, step-by-step derivation

1. The problem

Consider the one-dimensional heat equation on a rod of length L :

$$u_t = \kappa u_{xx}, \quad 0 < x < L, \quad t > 0.$$

Assume the ends of the rod are held at zero temperature (Dirichlet boundary conditions):

$$u(0,t) = 0, \quad u(L,t) = 0.$$

Finally, suppose the initial temperature distribution is prescribed:

$$u(x,0) = f(x).$$

Goal

We want to find $u(x,t)$. The strategy is to turn the PDE into ordinary differential equations, solve the resulting eigenvalue problem, and then use a Fourier sine series to match the initial condition.

2. Assume a separable solution

We look for special solutions in which the x -dependence and t -dependence are separated:

$$u(x,t) = X(x)T(t).$$

Then the derivatives are

$$u_t = X(x)T'(t), \quad u_{xx} = X''(x)T(t).$$

Substitute these into the heat equation:

$$XT' = \kappa X''T.$$

Divide by κXT :

$$T'/(\kappa T) = X''/X.$$

The left side depends only on t , while the right side depends only on x . For this equality to hold for every x and t , both sides must equal the same constant. We choose to write that constant as $-\lambda$:

$$T'/(\kappa T) = X''/X = -\lambda.$$

3. The two ODEs

Separating the equations gives

$$X'' + \lambda X = 0$$

$$T' + \kappa \lambda T = 0.$$

The spatial equation, together with the boundary conditions, is an eigenvalue problem. The boundary conditions determine which values of λ are allowed.

4. Apply the boundary conditions to the spatial equation

Since $u(x,t)=X(x)T(t)$, the conditions $u(0,t)=u(L,t)=0$ imply, for a nontrivial solution,

$$X(0)=0, \quad X(L)=0.$$

We now determine the allowed eigenvalues by considering the three possibilities for λ .

Case I: $\lambda = 0$

$$X'' = 0 \Rightarrow X = A + Bx.$$

$X(0)=0$ gives $A=0$. Then $X(L)=0$ gives $B=0$. Thus $X=0$, the trivial solution. Therefore $\lambda=0$ is not an eigenvalue.

Case II: $\lambda > 0$

$$\text{Let } \lambda = k^2. \text{ Then } X'' + k^2 X = 0.$$

$$X(x) = A \cos(kx) + B \sin(kx).$$

Applying $X(0)=0$ gives $A=0$, so $X=B \sin(kx)$. Applying $X(L)=0$ gives

$$B \sin(kL)=0.$$

For a nontrivial solution $B \neq 0$, so $\sin(kL)=0$. Hence

$$kL = n\pi, \quad n=1,2,3,\dots$$

Therefore the eigenvalues and eigenfunctions are

$$\lambda_n = (n\pi/L)^2$$

$$X_n(x) = \sin(n\pi x/L), \quad n=1,2,3,\dots$$

Case III: $\lambda < 0$

Write $\lambda = -\mu^2$. Then $X'' - \mu^2 X = 0$, whose solutions are combinations of $e^{\mu x}$ and $e^{-\mu x}$ (or $\cosh(\mu x)$ and $\sinh(\mu x)$). Applying $X(0)=X(L)=0$ again gives only the trivial solution. Thus there are no negative eigenvalues.

5. Solve the time equation

For each allowed eigenvalue $\lambda_n = (n\pi/L)^2$, the time equation is

$$T'_n + \kappa(n\pi/L)^2 T_n = 0.$$

Therefore

$$T_n(t) = \exp[-\kappa(n\pi/L)^2 t].$$

A separated solution for each mode is therefore

$$u_n(x,t) = C_n \sin(n\pi x/L) \exp[-\kappa(n\pi/L)^2 t].$$

6. Build the general solution

Because the heat equation is linear, we can add the individual separated solutions. This gives the general solution satisfying the boundary conditions:

$$u(x,t) = \sum_{n=1}^{\infty} C_n \sin(n\pi x/L) \exp[-\kappa(n\pi/L)^2 t].$$

7. Use the initial condition

Now we impose $u(x,0)=f(x)$. Setting $t=0$ makes every exponential factor equal to 1:

$$f(x) = \sum_{n=1}^{\infty} C_n \sin(n\pi x/L).$$

This is a Fourier sine series for the initial condition. The coefficients C_n are therefore the Fourier sine coefficients of $f(x)$.

8. Derive the coefficients using orthogonality

Multiply the Fourier series by $\sin(m\pi x/L)$ and integrate from 0 to L:

$$\int_0^L f(x) \sin(m\pi x/L) dx = \sum_{n=1}^{\infty} C_n \int_0^L \sin(n\pi x/L) \sin(m\pi x/L) dx.$$

The sine eigenfunctions are orthogonal:

$$\int_0^L \sin(n\pi x/L) \sin(m\pi x/L) dx = 0, \quad n \neq m,$$

while for $n=m$ the integral equals $L/2$. Thus every term disappears except the m th term:

$$\int_0^L f(x) \sin(m\pi x/L) dx = C_m L/2.$$

Solving for C_m gives

$$C_m = (2/L) \int_0^L f(x) \sin(m\pi x/L) dx.$$

Renaming the index $m \rightarrow n$, the standard result is

$$C_n = (2/L) \int_0^L f(x) \sin(n\pi x/L) dx.$$

9. Final solution

Once the coefficients have been calculated from the initial condition, substitute them into

$$u(x,t) = \sum_{n=1}^{\infty} C_n \sin(n\pi x/L) \exp[-\kappa(n\pi/L)^2 t].$$

10. The conceptual picture

Boundary conditions → **allowed spatial modes.**

The boundary conditions determine the eigenvalues λ_n and eigenfunctions X_n .

Initial condition → **amount of each mode.**

The initial function $f(x)$ determines C_n , the amount of each spatial eigenfunction present.

Time equation → **evolution of each mode.**

Each mode decays as $\exp[-\kappa(n\pi/L)^2 t]$. Higher-frequency modes decay faster.

The overall chain is therefore:

PDE → separation → spatial eigenvalue problem → boundary conditions → eigenfunctions → superposition →
Fourier expansion of the initial condition → time evolution.

11. Connection to Sturm–Liouville theory

The spatial equation can be written as $-X'' = \lambda X$. This is a Sturm–Liouville problem with $p(x)=1$, $q(x)=0$, and weight $w(x)=1$. Its eigenfunctions are orthogonal, which is exactly why the Fourier coefficients can be extracted by projection. Thus Fourier series are a special case of the more general Sturm–Liouville eigenfunction-expansion method.